MATGEO Presentation: 4.3.44

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Problem Statement

Find the coordinates of the point where the line through $(5, 1, 6)$ and $(3, 4, 1)$ crosses the YZ-plane.

Given data

The two points on the line are given by the position vectors:

$$\mathbf{A} = \begin{pmatrix} 5 \\ 1 \\ 6 \end{pmatrix} \tag{1}$$

$$\mathbf{B} = \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} \tag{2}$$

Formulae

The vector equation of a line passing through points **A** and **B** is given by:

$$\mathbf{x} = \mathbf{A} + k\left(\mathbf{B} - \mathbf{A}\right) \tag{3}$$

where k is a scalar parameter.
The equation of the YZ-plane is $x = 0$. A point $\mathbf{x}$ lies on this plane if its x-component is zero. This can be expressed as:

$$\mathbf{e_1}^\top \mathbf{x} = 0 \tag{4}$$

where $\mathbf{e_1} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

Solving

First, find the direction vector of the line:

$$\mathbf{B} - \mathbf{A} = \begin{pmatrix} 3-5 \\ 4-1 \\ 1-6 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \\ -5 \end{pmatrix} \tag{5}$$

The parametric equation of the line is:

$$\mathbf{x} = \begin{pmatrix} 5 \\ 1 \\ 6 \end{pmatrix} + k \begin{pmatrix} -2 \\ 3 \\ -5 \end{pmatrix} = \begin{pmatrix} 5-2k \\ 1+3k \\ 6-5k \end{pmatrix} \tag{6}$$

To find the intersection with the YZ-plane, we set the x-component to zero:

$$5 - 2k = 0 \implies 2k = 5 \implies k = \frac{5}{2} \tag{7}$$

Substitute $k = 5/2$ back into the line equation to find the point:

$$\mathbf{x} = \begin{pmatrix} 5-2(5/2) \\ 1+3(5/2) \\ 6-5(5/2) \end{pmatrix} = \begin{pmatrix} 0 \\ 1+15/2 \\ 6-25/2 \end{pmatrix} = \begin{pmatrix} 0 \\ 17/2 \\ -13/2 \end{pmatrix} \tag{8}$$

Plot